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Acetylcysteine Powder for Oral Solution 200 mg

Product Name

ACEVIZ (Acetylcysteine Powder for Oral Solution 200 mg)

Name and Strength of Active Substance(s)

Acetylcysteine 200 mg

Product Description

White to off-white, granular powder after reconstitution in warm water forms a clear solution.

Pharmacodynamics & Pharmacokinetics

Pharmacodynamics

Pharmacotherapeutic group: Mycolytics, ATC code: R05CB01
N-acetyl-L-cystineine (NAC), the active ingredient in Acetylcysteine 200 mg Powder for Oral Solution exerts an intense mucolytic-fluidizing action on mucous and mucopurulent secretions by depolymerizing the mucoproteic complexes and the nucleic acids which confer viscosity to the vitreous and purulent component of the sputum and other secretions. Furthermore, acetylcysteine exerts a direct antioxidant action, having a free thiol (- SH) nucleophilic group that is able to interact directly with electrophilic groups of oxidant radicals. Of particular interest is the recent finding that acetylcysteine protects α1-antitrypsin enzyme inhibiting elastase from inactivation by hypochlorous acid (HOCl), a powerful oxidant agent produced by the myeloperoxidase enzyme of activated phagocytes. Due to its molecular structure, acetylcysteine can readily cross cell membranes. Inside the cell, NAC is deacetylated to L-cysteine, an amino acid essential for glutathione synthesis (GSH). GSH is a highly reactive tripeptide found ubiquitously in the various tissues of animals and is essential for the maintenance of functional capacity as well as cellular morphological integrity. It is the most important protective intracellular mechanism against oxidant radicals, both exogenous and endogenous, as well as toward numerous cytotoxic substances. These features make Acetylcysteine 200 mg Powder for Oral Solution particularly suitable for the treatment of acute and chronic affections of the respiratory system, characterised by thick, viscous mucous and mucopurulent secretions. There is no evidence on the efficacy and safety of mucolytic s including acetylcysteine in acute bronchitis.

Pharmacokinetics

Absorption

Following oral administration, acetylcysteine is rapidly and almost completely absorbed and metabolised in the liver to cysteine (the pharmacologically active metabolite), diacetylcysteine, cysteine and further mixed disulphides.

Distribution

Due to the high first-pass effect, the bioavailability of orally administered acetylcysteine is very low (approximately 10%). In humans, maximum plasma concentrations are achieved

after 1-3 hours with the maximum plasma concentration of the metabolite cysteine in the range of approximately 2µmol/l. The protein binding of Acetylcysteine was determined to be about 50%.
Biotransformation
Acetylcysteine and its metabolites occur in three different forms in the organism: partially in free form, partially bound to proteins via labile disulphide bonds and partially as incorporated amino acid. Acetylcysteine is excreted almost exclusively in the form of inactive metabolites (inorganic sulphates, diacetylcysteine) via the kidneys. The plasma half-life of Acetylcysteine is approximately 1 hour and is mainly determined by the rapid hepatic biotransformation. Impaired hepatic function therefore leads to prolonged plasma half-lives of up to 8 hours.
Elimination
Pharmacokinetic studies with intravenous administration of acetylcysteine revealed a distribution volume of 0.47 l/kg (in total) or 0.59 l/kg (reduced acetylcysteine); the plasma clearance was determined to be 0.11 l/h/kg (in total) and 0.84 l/h/kg (reduced acetylcysteine), respectively. The elimination half-life after intravenous administration is 30-40 minutes while excretion follows three-phase kinetics (alpha, beta and terminal gamma phase).
Acetylcysteine crosses the placenta and is detected in cord blood. No information is available regarding excretion in breast milk.No knowledge is available concerning the behaviour of acetylcysteine at the blood- brain barrier in humans

Indication

All respiratory tract diseases leading to the formation of thick secretions difficult to be expectorated, such as acute and chronic bronchitis, laryngitis, sinusitis, tracheitis, influenza, bronchial asthma and (as complementary treatment) mucoviscidosis.

Recommended Dosage

Usual posology for acute diseases:
Adults & children above 12 years old: 600 mg daily, divided into one or more administrations (e.g. 200 mg sachets 3 times daily).
Children from 2 to 12 years of age: 200 mg twice daily.
Special posologies:
Long-term treatment: 400–600 mg daily, divided into one or more administrations, with a maximum treatment duration of 3 - 6 months.
If the excessive mucus production and the consequent cough do not disappear after a two- week treatment, the diagnosis should be re-evaluated in order to exclude a possible malignant disease of the respiratory tract.
Route of administration: Oral
Method of administration
Dissolve the contents of one sachet completely in a glass containing a little water just before use, stirring as needed with a teaspoon.

Contraindications

Hypersensitivity to the active substance or to any of the excipients.
Contraindicated in children below two (2) years of age.

Warnings and Precautions

JAUHKAN UBAT DARIPADA CAPAIAN KANAK-KANAK
‘KEEP OUT OF REACH OF CHILDREN’
Please consult your pharmacist/doctor before taking this product.

Patients with bronchial asthma should be closely monitored during therapy; if bronchospasm occurs, treatment with Acetylcysteine Powder for Oral Solution should be discontinued immediately.
Administration of acetylcysteine, especially at the beginning of treatment, may liquefy bronchial secretions and, at the same time, increase their volume. If the patient is unable to expectorate efficiently, to avoid retention of secretions postural drainage and tracheal suction should be used.
Acetylcysteine can cause interference with the colorimetric assay method for the determination of salicylates.
Acetylcysteine can interfere with tests for ketones in urine.

Interactions with Other Medicaments

Antitussive drugs and acetylcysteine should not be administered concomitantly because reducing the cough reflex may lead to a build-up of bronchial secretions.
Activated charcoal may reduce the effect of acetylcysteine.
It is advisable not to mix Acetylcysteine Powder for Oral Solution with other medicinal products.
In vitro tests have shown that when cephalosporin antibiotics and acetylcysteine are mixed, there is a degree of antibiotic inactivation. It is precautionary to advise the administration of oral antibiotics at least two hours before or after acetylcysteine.
Concurrent administration of nitroglycerin and acetylcysteine causes significant hypotension and leads to temporal artery dilation with possible onset of headache.
If concurrent administration of nitroglycerin and acetylcysteine is required, patients should be monitored and warned for hypotension that can be severe and accompanied by a headache.

Statement on usage during pregnancy and lactation

Pregnancy

Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. As a precautionary measure, it is preferable to avoid the use of Acetylcysteine Powder for Oral Solution during pregnancy.

Lactation

There is insufficient information on the excretion of acetylcysteine in human milk. A risk to the newborns/infants cannot be excluded.

Side Effects

Immune System Disorders:

Anaphylactic / anaphylactoid reaction

Skin and Subcutaneous Tissue Disorders:

Severe cutaneous adverse reactions (SCAR) e.g. erythema multiforme, Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). In most of these cases reported atleast one other drug was administered at the same time, which may have possibly enhanced the described mucocutaneous effects.
Adverse reactions are listed below, by system organ class and frequency.

	Frequency / Adverse Reactions			
System Organ Class	Uncommon (≥1/1,000, < 1/100)	Rare (≥1/10,000, < 1/1,000)	Very rare (< 1/10,000)	Not Known

Immune system disorders	Hypersensitivity	Anaphylactic shock, anaphylactic/anaphylactoid reaction
Nervous system disorders	Headache	
Ear and labyrinth disorders	Tinnitus	
Cardiac disorders	Tachycardia	
Vascular disorders	Hypotension	Haemorrhage
Respiratory, thoracic and mediastinal disorders	Hypotension	Bronchospasm, dyspnoea
Gastrointestinal disorders	Vomiting, diarrhoea, stomatitis, abdominal pain, nausea	Dyspepsia
Skin and subcutaneous tissue disorders	Urticaria, rash, angioedema, pruritus	
General disorders and administration site conditions	Fever	Oedema of the face

In case of recurrence skin and mucosal lesions, medical advice should be sought at once and the use of acetylcysteine terminated immediately.
A decreased blood platelet aggregation in the presence of acetylcysteine has been confirmed by various studies. The clinical relevance has not yet been clarified to date.

Overdose and Treatment

An acute overdose of acetylcysteine can cause gastrointestinal symptoms such as nausea, vomiting and diarrhoea.

Treatment of Overdose

Treatment of overdose is to be symptomatic and supportive treatment as indicated by the patient's clinical condition.

Storage Conditions

Store at temperature not exceeding 30°C.
This product should be used as a single use.
Jauhkan daripada capaian kanak-kanak.
Keep medicines out of reach of children.

Dosage forms and packaging available

Powder for oral solution
30 & 50 sachets in a carton.

Date of Revision :

11/03/2020

Manufactured by :
XL LABORATORIES PVT. LTD.
E-1223, Phase-I Extn., (Ghatat) RLICO Indl. Area, Bhiwadi-301019, Rajasthan, India

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PLO1254-00

Brand Name with strength: ACEVIZ (Acetylcistiene 200mg Powder for Oral Solution)					Country/Market: Malaysia		New Artwork
Leaf insert A/W Code: PL01254-00		Leaf insert Pharma Code: 104240		Leaf insert Size: 165x120mm		Date: 10/04/2026	
Carton Size: 115x50x85mm			Pack Size: 30 Sachet		Colour: XXXXXXXXXX Black 100%		
Date	Design		Production		QC	QA	RA
Sign							
	<i>Prepared By</i>		<i>Checked By</i>		<i>Checked By</i>	<i>Approved By</i>	<i>Checked By</i>

PL01254-00